

Ideas

Self Balancing 2-Wheeled Robot



Use Gyroscope + Accelerometer to measure pitch angle, use a filter to estimate the real value since gyro is noisy.

Use PID to control a PWM signal to supply the

two motors.

$$K_p = 414,8$$

$$K_i = 2449$$

$$K_d = 17,56$$

$$\ddot{x} = \frac{[-\sin\theta \cdot \dot{\theta}^2 \cdot l^3 \cdot m^2 - \sin\theta \cdot I \cdot \dot{\theta}^2 \cdot l \cdot m + g \cdot (\cos\theta \cdot \sin\theta \cdot l^2 \cdot m^2 + ulm + uI)]}{[I \cdot m + I \cdot M + l^2 \cdot m^2 - l^2 \cdot m^2 \cdot \cos^2\theta + M \cdot l^2 \cdot m]}$$

$$\ddot{\theta} = \frac{[l \cdot m \cdot (-l \cdot m \cdot \cos\theta \cdot \sin\theta \cdot \dot{\theta}^2 + u \cdot \cos\theta + g \cdot m \cdot \sin\theta + M \cdot g \cdot \sin\theta)]}{[I \cdot m + I \cdot M + l^2 \cdot m^2 - l^2 \cdot m^2 \cdot \cos^2\theta + M \cdot l^2 \cdot m]}$$